16A 1 STANDARD ELECTRODE (REDOX) POTENTIALS

SPECIFICATION REFERENCE
16.1 16.2 16.3(i) 16.3(ii) 16.3(iii) 16.4 16.5(i) 16.5(ii) 16.13

LEARNING OBJECTIVES

- Understand the terms 'oxidation' and 'reduction' in terms of electron transfer and changes in oxidation number, applied to *s*–, *p*– and *d*–block elements.
- Know what is meant by the term 'standard electrode potential', $E^{\ominus}$.
- Know that the standard electrode potential, $E^{\ominus}$, is measured in conditions of:
  - (i) 298 K temperature
  - (ii) 100 kPa pressure of gases
  - (iii) 1.00 mol dm^{-3} concentration of ions.
- Know the features of the standard hydrogen electrode and understand why a reference electrode is necessary.
- Understand that different methods are used to measure standard electrode potentials of:
  - (i) metals or non-metals in contact with their ions in aqueous solution
  - (ii) ions of the same element with different oxidation numbers.
- Know that standard electrode potentials are sometimes referred to as standard redox potentials and can be listed as an electrochemical series.

REVISION OF OXIDATION AND REDUCTION

You should remember from **Topic 8 (Book 1: IAS)** that there are two very useful definitions of oxidation and reduction. One is in terms of loss or gain of electrons. The other is in terms of changes to the oxidation number of an element.

In terms of electrons:

- Oxidation is the loss of electrons.
- Reduction is the gain of electrons.

In terms of changes in oxidation number:

- An element in a species is oxidised when its oxidation number increases.
- An element in a species is reduced when its oxidation number decreases.

These definitions apply to all elements whether they are in the s-, p- or d-block of the Periodic Table.

How well you will understand the topic of electrode potentials depends very heavily on your appreciation of redox reactions. If you are at all unsure of your understanding, then we suggest that you revisit **Topic 8 (Book 1: IAS)**.

STANDARD ELECTRODE (REDOX) POTENTIAL

The two terms 'standard electrode potential' and 'standard redox potential' mean the same thing. The reactions involved in the measurement of a standard electrode potential are redox reactions. Therefore, the term redox potential is also commonly used. In order to match the Edexcel specification we will use the term 'standard electrode potential' throughout this book.

There are many ways of teaching this topic, but very often a crucial mistake is made at the outset. It is sometimes forgotten to emphasise that the reactions involved are *equilibria*. Too often you will find that the equations are written as if they were one-way reactions, rather than reversible. This very simple mistake can make the whole topic much more difficult to understand than is necessary.

Throughout this topic we will avoid this problem by always talking in terms of equilibria.

BACKGROUND: THE DIFFERENT TENDENCIES OF METALS TO RELEASE ELECTRONS TO FORM POSITIVE IONS

When a metal such as magnesium or copper is placed in water, there is a very small tendency for the metal atoms to lose electrons and go into solution as positive ions:

$$M(s) \rightarrow M^{n+}(aq) + ne^- \text{ (where M = Mg or Cu)}$$

The electrons will remain on the surface of the metal.

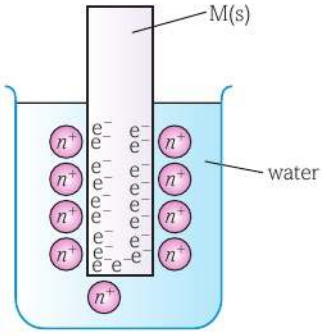
In a very short time there will be a build-up of electrons on the surface of the metal, and the resulting negative charge attracts positive ions. In this way, a layer of positive ions is formed surrounding the metal (**fig A**).

Some of the positive ions will regain their electrons from the surface of the metal and return to form part of the metal surface:

$$M^{n+}(aq) + ne^- \rightarrow M(s)$$

Eventually, a dynamic equilibrium will be established in which the rate at which ions are leaving the surface of the metal to go into solution is the same as the rate at which they are joining it from solution. This equilibrium is represented by the equation:

$$M^{n+}(aq) + ne^- \rightleftharpoons M(s)$$



▲ **fig A** Metal M immersed in water. n^+ represents metal ions.

The equations for magnesium and copper are:

$$Mg^{2+}(aq) + 2e^- \rightleftharpoons Mg(s)$$

$$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$$

The difference between magnesium and copper is that the equilibrium position will be further to the left-hand side for

magnesium than for copper, because the tendency of magnesium to release electrons is greater than that of copper. So, with magnesium there will be a greater negative charge on the metal, and more positive ions in solution.

LEARNING TIP

Note how the two equilibria are written. By convention they are written with the electrons on the *left*-hand side. Make sure you stick to this convention at all times.

In the case of both magnesium and copper there is a potential difference between the metal and the solution. However, the potential difference is greater with magnesium than with copper. In each case, we would like to measure the potential difference between the metal and the solution. This is called the **absolute potential difference**.

However, it is not possible to measure the absolute potential difference between a metal electrode and its solution. The reason is that although it is a simple matter to connect the metal electrode to one terminal of a voltmeter, the other terminal would have to be connected to the solution, and the only way of doing this is to dip another piece of metal into the solution. This second piece of metal would create its own potential difference. So, you would be measuring the potential difference between the two pieces of metal, and not between the original metal and the solution.

The way to solve this problem is to create a reference electrode and then measure the difference in potential between this reference electrode and the metal electrode.

STANDARD HYDROGEN ELECTRODE

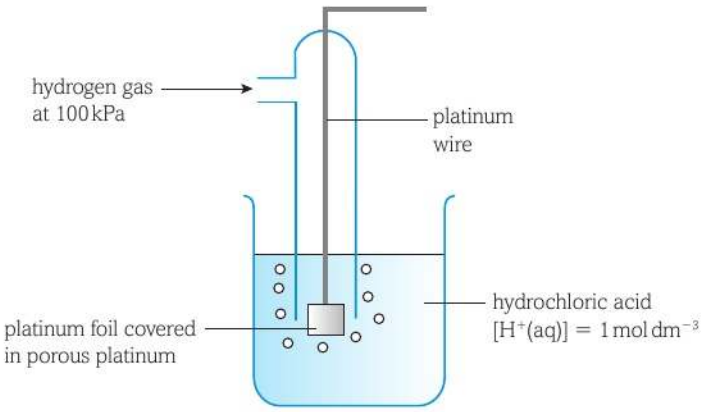
The reference electrode of choice is the standard hydrogen electrode. This electrode consists of hydrogen gas at a pressure of 100 kPa (1 bar) bubbling over a piece of platinum foil dipped into a solution of hydrochloric acid (or sulfuric acid) with a hydrogen ion concentration of 1 mol dm^{-3}, at a temperature of 298 K.

The surface of the platinum foil is covered in porous platinum. Porous platinum has a large surface area and allows an equilibrium between hydrogen ions in solution and hydrogen gas to be established quickly.

$$H^+(aq) + e^- \rightleftharpoons \tfrac{1}{2}H_2(g)$$

It is this equilibrium that we are going compare with all others.

Fig B shows a cross section of a typical standard hydrogen electrode.



▲ **fig B** A standard hydrogen electrode.

IMPORTANCE OF USING STANDARD CONDITIONS

You will remember that the position of an equilibrium can be changed by altering the conditions. If we are going to make fair comparisons it is therefore necessary to standardise the conditions used. The standard conditions are:

- gas pressure, 100 kPa (1 bar)
- temperature, 298 K
- concentration of ions in solution, 1 mol dm^{-3}.

These standard conditions apply to all equilibria.

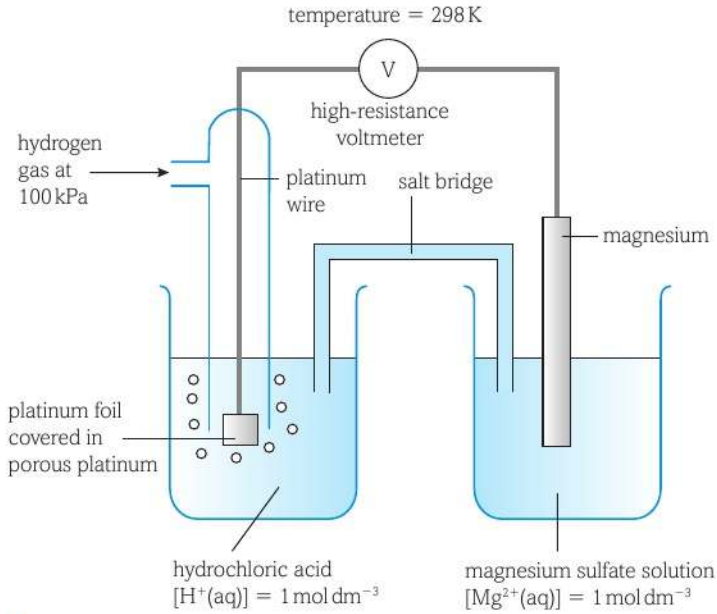
EXAM HINT

Many exam questions will ask you to provide a clear definition for standard electrode potential. This may be worth 2 or 3 marks. Make sure that you know the exact definition, including the specific standard conditions.

MEASURING A STANDARD ELECTRODE POTENTIAL

We are now in a position to appreciate how to measure the standard electrode potential of a metal ion | metal system, such as that between magnesium ions and magnesium (Mg^{2+} | Mg).

To do this we connect the standard hydrogen electrode to the magnesium electrode via a circuit containing a high resistance voltmeter. **Fig C** shows the arrangement of the apparatus.



▲ **fig C** Using a standard hydrogen electrode to measure a standard electrode potential.

The two components of the apparatus are known as half-cells. The two half-cells combine to make a complete cell.

The salt bridge is needed to complete the electrical circuit. It usually contains a concentrated solution of potassium nitrate in the form of liquid or a gel. It functions by allowing the movement of ions. Theoretically, the salt bridge can contain any ionic salt, but neither of the ions present should interfere with the components of the half-cells. For example, if the right-hand half-cell contains a solution of silver nitrate, then potassium chloride would be unsuitable for the salt bridge because the chloride ions would interact with the silver ions to form a precipitate of silver chloride.

DID YOU KNOW?

The best instrument to use to measure the potential difference is a potentiometer. No electrons flow when this instrument is used, so both half-cell reactions are in equilibrium.

However, the best practical way of measuring the potential difference is to use a high-resistance voltmeter.

There is a reason why a high-resistance voltmeter is used. Ideally, the voltmeter should have infinite resistance so that there is no flow of electrons (i.e. no current flowing) around the external circuit. If this were possible, the reading on the voltmeter would represent the difference in potential between the two half-cells when *both reactions are in equilibrium*. The two reactions concerned are:

$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g)$

$Mg^{2+}(aq) + 2e^- \rightleftharpoons Mg(s)$

With the magnesium half-cell connected to the standard hydrogen electrode, the reading on the voltmeter is 2.37 V.

If the magnesium half-cell is replaced by a copper half-cell (i.e. copper metal dipped into a solution of copper sulfate with $[Cu^{2+}(aq)] = 1\ mol\ dm^{-3}$), then the reading on the voltmeter would be 0.34 V.

There is, however, a further issue to be considered. In the case of magnesium, if the voltmeter were taken out of the circuit, electrons would flow from the magnesium electrode to the hydrogen electrode. This means that magnesium is the negative electrode of the cell.

With copper, electrons would flow from the hydrogen electrode to the copper electrode, making copper the positive electrode of the cell.

This difference in behaviour is recognised by using a sign convention. A negative sign (−) indicates that the metal electrode is negative with respect to the hydrogen electrode, while a positive sign indicates the reverse.

The difference in potential measured, together with the appropriate sign, is called the **standard electrode potential** of the metal. Standard electrode potential is given the symbol $E^\ominus$.

Standard electrode potentials are quoted together with the relevant half-cell reaction. For example, for magnesium and copper they are:

$Mg^{2+}(aq) + 2e^- \rightleftharpoons Mg(s) \quad E^\ominus = -2.37\ V$

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s) \quad E^\ominus = +0.34\ V$

By convention, the standard electrode potential of the standard hydrogen electrode is zero.

$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g) \quad E^\ominus = 0.00\ V$

Some other $E^\ominus$ values, together with the relevant half-cell reactions, are shown in **table A**.

HALF-CELL REACTION	$E^\ominus$/V
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.80
$Al^{3+}(aq) + 3e^- \rightleftharpoons Al(s)$	−1.66
$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s)$	−0.74
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.44
$Ni^{2+}(aq) + 2e^- \rightleftharpoons Ni(s)$	−0.25
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76

table A

LEARNING TIP

The sign of a standard electrode potential indicates the polarity of the electrode when compared with the hydrogen electrode. This sign is fixed and does not change if the equation for the half-cell reaction is reversed, as sometimes stated elsewhere. Remember, standard electrode potentials are measured when *no electrons are flowing* and hence both half-cell reactions are *in equilibrium*.

So, if the half-cell reaction for magnesium is represented as: $Mg(s) \rightleftharpoons Mg^{2+}(aq) + 2e^-$, the $E^\ominus$ value is still −2.37 V. It does not change to +2.37 V.

Standard electrode potential is described as being a '*sign invariant quantity*'. This means that however the reaction is represented, the sign *remains the same*.

LEARNING TIP

Remember, a negative $E^\ominus$ does not suggest that the equilibrium of the half-cell reaction lies to the left. It merely means that it lies *further to the left* than the equilibrium of the standard hydrogen electrode.

In the same way, a positive $E^\ominus$ means that the equilibrium of the half-cell reaction lies *further to the right* than the equilibrium of the standard hydrogen electrode.

WHAT DO $E^\ominus$ VALUES TELL US?

It is important to remember that the standard electrode potential of a metal ion | metal half-cell is the potential difference measured when the half-cell is connected to a standard hydrogen electrode.

The potential difference provides a *comparison* between the position of the metal ion | metal equilibrium and the position of equilibrium in the hydrogen electrode.

To illustrate this point, let us consider the following two standard electrode potentials:

$Mg^{2+}(aq) + 2e^- \rightleftharpoons Mg(s) \quad E^\ominus = -2.37\ V$

$H^+(aq) + 2e^- \rightleftharpoons \frac{1}{2}H_2(g) \quad E^\ominus = 0.00\ V$

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s) \quad E^\ominus = +0.34\ V$

The negative sign for $Mg^{2+}(aq) \mid Mg(s)$ indicates that the equilibrium position of this reaction is further to the left than the equilibrium position of the reaction in the hydrogen electrode.

The positive sign for $Cu^{2+}(aq) \mid Cu(s)$ indicates that the equilibrium position of this reaction is further to the right than the equilibrium position of the reaction in the hydrogen electrode.

This is another way of saying that magnesium releases electrons more readily than hydrogen (and also than copper), but that copper releases electrons less readily than hydrogen.

Magnesium is, therefore, a better reducing agent than both hydrogen and copper. In addition, hydrogen is a better reducing agent than copper. This point will be developed later in the topic.

LEARNING TIP

A reducing agent is a species that reduces another species by adding one or more electrons to it. When a reducing agent reacts it loses electrons and is, therefore, oxidised.

An oxidising agent is a species that oxidises another species by removing one or more electrons from it. When an oxidising agent reacts it gains electrons and is, therefore, reduced.

EXAM HINT

It is useful to remember that the electrode where oxidation takes place is the anode. This is true for all electrochemical and electrolytic cells.

SUMMARY

$E^\ominus$ values provide a method of comparing the positions of equilibria when metal atoms lose electrons to form ions in solution.

- The more negative the $E^\ominus$ value, the further the equilibrium lies to the left, i.e. the more readily the metal loses electrons to form ions.
- The more positive (or less negative) the $E^\ominus$ value, the further the equilibrium lies to the right, i.e. the less readily the metal loses electrons to form ions.

Later in this topic we will extend this to include both molecules and ions losing electrons.

ELECTROMOTIVE FORCE

As already mentioned, when we are determining a standard electrode potential, the potential difference is measured when *no electrons are flowing* (i.e. no current is flowing) through the external circuit. The potential difference measured under these conditions is called the **electromotive force (emf)** of the cell. The standard emf of a cell is given the symbol $E^\ominus_{cell}$ and is sometimes called the standard cell potential.

Emf values can be positive or negative, depending on the reference point against which the potential difference is measured. When measuring standard electrode potentials, the reference point is the potential of the standard hydrogen electrode, which is set at zero. The emf of a cell in which the hydrogen electrode is the positive electrode will have a negative value. Similarly, the emf of a cell in which the hydrogen electrode is the negative electrode will have a positive value.

This now allows us to define standard electrode potential as follows:

The standard electrode potential of a half-cell is the emf of a cell containing the half-cell connected to the standard hydrogen electrode. Standard conditions of 298 K, 100 kPa pressure of gases and solution concentrations of 1 mol dm^{-3} apply.

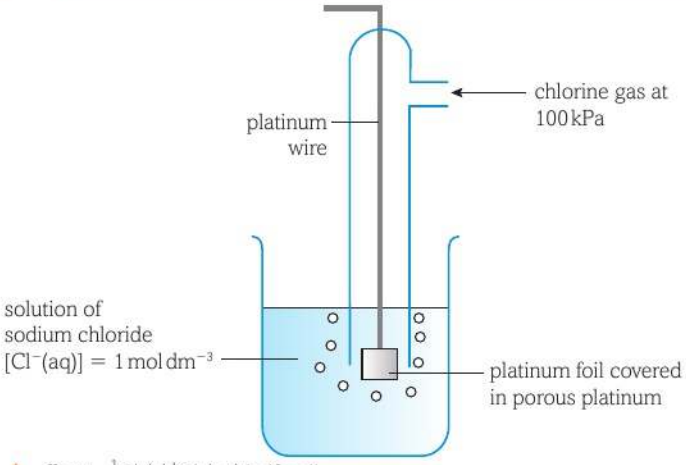
MEASURING STANDARD ELECTRODE POTENTIALS OF MORE COMPLICATED REDOX SYSTEMS

SYSTEMS INVOLVING GASES

Suppose we wanted to measure the standard electrode potential of the following redox system:

$$\tfrac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq)$$

We can do this by setting up a half-cell in which chlorine gas is bubbled into a solution containing chloride ions. However, in order to establish an equilibrium between the chlorine molecules and the chloride ions, and also to provide an electrical connection to the external circuit, we would need to place a piece of platinum into the solution. **Fig D** shows the set up.



▲ **fig D** $\frac{1}{2}Cl_2(g)|Cl^-(aq)$ half-cell.

This half-cell is then connected to a standard hydrogen electrode and the emf is measured in the usual way. The measured emf is +1.36 V. Hence:

$$\frac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq) \qquad E^\ominus = +1.36\ V$$

OTHER SYSTEMS

1. Non-metal elements and their ions in solution

Examples: $\frac{1}{2}Br_2(aq) \mid Br^-(aq)$ and $\frac{1}{2}I_2(aq) \mid I^-(aq)$

To determine the $E^\ominus$ of the redox system $\frac{1}{2}Br_2 \mid Br^-$, a half-cell containing a solution of Br_2 and Br^- ions, each of concentration 1 mol dm^{-3}, is connected to a standard hydrogen electrode. A similar set-up is used to determine the $E^\ominus$ of the redox system $\frac{1}{2}I_2 \mid I^-$.

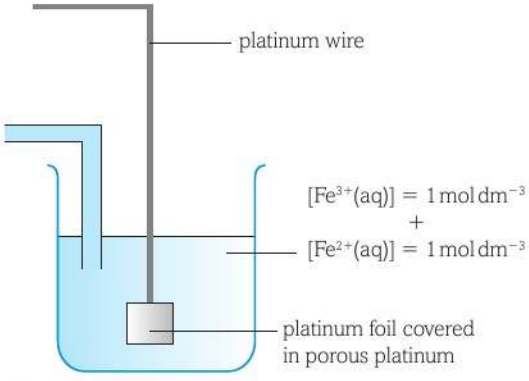
$$\frac{1}{2}Br_2(aq) + e^- \rightleftharpoons Br^-(aq) \qquad E^\ominus = +1.09\ V$$

$$\frac{1}{2}I_2(aq) + e^- \rightleftharpoons I^-(aq) \qquad E^\ominus = +0.54\ V$$

2. Ions of the same element with different oxidation numbers

Example: $Fe^{3+}(aq) \mid Fe^{2+}(aq)$

The half-cell is arranged as shown in **fig E**.



▲ **fig E** $Fe^{3+}(aq)|Fe^{2+}(aq)$ half-cell.

Note that each ion has a concentration of 1 mol dm^{-3}.

$$Fe^{3+}(aq) + e^- \rightleftharpoons Fe^{2+}(aq) \qquad E^\ominus = +0.77\ V$$

THE ELECTROCHEMICAL SERIES

ARRANGING REDOX EQUILIBRIA IN ORDER OF THEIR $E^\ominus$ VALUES

The electrochemical series is built up by arranging various redox equilibria in order of their standard electrode (redox) potentials. The most negative $E^\ominus$ values are placed at the top of the series, and the most positive at the bottom.

Table B shows part of the electrochemical series.

HALF-CELL REACTION	$E^\ominus$/V
$Li^+(aq) + e^- \rightleftharpoons Li(s)$	−3.03
$K^+(aq) + e^- \rightleftharpoons K(s)$	−2.92
$Ca^{2+}(aq) + 2e^- \rightleftharpoons Ca(s)$	−2.87
$Na^+(aq) + e^- \rightleftharpoons Na(s)$	−2.71
$Al^{3+}(aq) + 3e^- \rightleftharpoons Al(s)$	−1.66
$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$	−0.76
$Cr^{3+}(aq) + 3e^- \rightleftharpoons Cr(s)$	−0.74
$Fe^{2+}(aq) + 2e^- \rightleftharpoons Fe(s)$	−0.44
$H^+(aq) + e^- \rightleftharpoons \frac{1}{2}H_2(g)$	0.00
$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$	+0.34
$\frac{1}{2}I_2(aq) + e^- \rightleftharpoons I^-(aq)$	+0.54
$Ag^+(aq) + e^- \rightleftharpoons Ag(s)$	+0.80
$\frac{1}{2}Br_2(aq) + e^- \rightleftharpoons Br^-(aq)$	+1.09
$\frac{1}{2}Cl_2(aq) + e^- \rightleftharpoons Cl^-(aq)$	+1.36
$\frac{1}{2}F_2(aq) + e^- \rightleftharpoons F^-(aq)$	+2.87

table B

REDUCING AGENTS AND OXIDISING AGENTS

The species on the right-hand side of the half-cell reactions are all capable of behaving as reducing agents because they can lose electrons.

The most powerful reducing agent in **table B** is lithium because its redox system has the most negative $E^\ominus$ value; the equilibrium position of its half-cell reaction is furthest to the left.

The least powerful reducing agent in the list is the fluoride ion, $F^-(aq)$ because its redox system has the least negative (most positive) $E^\ominus$ value. The position of equilibrium of its half-cell reaction is furthest to the right.

The species on the left-hand side of the half-cell reactions are all capable of acting as oxidising agents because they can gain electrons.

The most powerful oxidising agent is fluorine and the least powerful is the lithium ion, $Li^+(aq)$.

Table C summarises these statements.

	OXIDISED FORM	REDUCED FORM		$E^{\ominus}$/V
least powerful oxidising agent ↓	$Li^+(aq)$	$Li(s)$	*most powerful reducing agent* ↑	−3.03
	$K^+(aq)$	$K(s)$		−2.92
	$Ca^{2+}(aq)$	$Ca(s)$		−2.87
	$Na^+(aq)$	$Na(s)$		−2.71
	$Al^{3+}(aq)$	$Al(s)$		−1.66
	$Zn^{2+}(aq)$	$Zn(s)$		−0.76
	$Cr^{3+}(aq)$	$Cr(s)$		−0.74
	$Fe^{2+}(aq)$	$Fe(s)$		−0.44
	$H^+(aq)$	$\frac{1}{2}H_2(g)$		0.00
	$Cu^{2+}(aq)$	$Cu(s)$		+0.34
	$\frac{1}{2}I_2(aq)$	$I^-(aq)$		+0.54
	$Ag^+(aq)$	$Ag(s)$		+0.80
	$\frac{1}{2}Br_2(aq)$	$Br^-(aq)$		+1.09
most powerful oxidising agent	$\frac{1}{2}Cl_2(aq)$	$Cl^-(aq)$	*least powerful reducing agent*	+1.36
	$\frac{1}{2}F_2(aq)$	$F^-(aq)$		+2.87

table C

DID YOU KNOW?

The standard electrode potentials of lithium, sodium, calcium and potassium cannot be determined experimentally because each of these metals reacts to completion with water; i.e. they do not establish an equilibrium. The $E^{\ominus}$ values for these metals are calculated from thermodynamic data.

The same is true for fluorine, which also reacts with water.

SUMMARY

The more negative the $E^{\ominus}$ value, the more the equilibrium position lies towards the left – and the more readily the species on the right loses electrons. The more negative the $E^{\ominus}$ value, the more powerful the reducing agent.

The more positive the $E^{\ominus}$ value, the more the equilibrium position lies towards the right – and the more readily the species on the left loses electrons. The more positive the $E^{\ominus}$ value, the more powerful the oxidising agent.

CHECKPOINT

SKILLS CREATIVITY

1. Draw a labelled diagram of apparatus that could be used to measure the standard electrode potential of a $Cu^{2+}(aq)|Cu(s)$ half-cell.
2. The standard electrode potential of the $\frac{1}{2}Br_2(aq)|Br^-(aq)$ half-cell is +1.09 V. Explain what is meant by this statement and state the significance of the positive sign.
3. What is meant by the term 'electromotive force (emf)'. How is the electromotive force of a cell measured?

SUBJECT VOCABULARY

absolute potential difference the potential difference between a metal and a solution of its ions
standard electrode potential the emf measured when a half-cell is connected to a standard hydrogen electrode under standard conditions
electromotive force (emf) the measured potential difference of a cell when no current is flowing

16A 2 ELECTROCHEMICAL CELLS

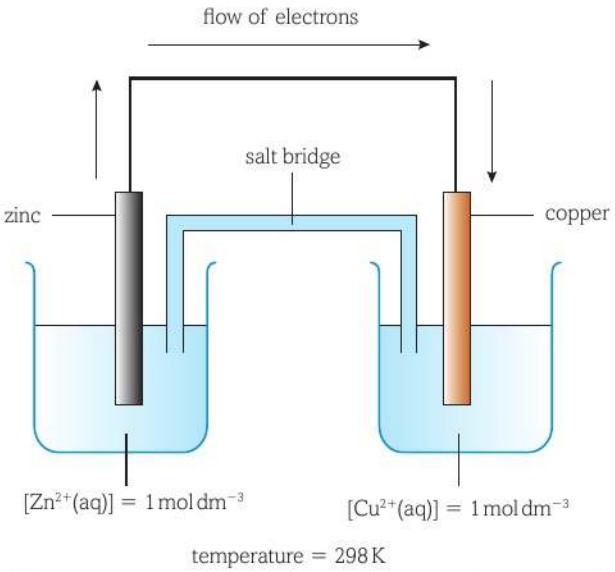
SPECIFICATION REFERENCE
16.7 16.8
CP12

LEARNING OBJECTIVES

- Be able to calculate a standard emf, $E^{\ominus}_{cell}$, by combining two standard electrode potentials.
- Be able to write cell diagrams using the conventional representation of half-cells.

ELECTROCHEMICAL CELLS

An electrochemical cell is a device for producing an electric current from chemical reactions. It is constructed from two half-cells. **Fig A** shows the apparatus used to construct an electrochemical cell from a Zn^{2+} | Zn half-cell and a Cu^{2+} | Cu half-cell under standard conditions.



▲ **fig A** An electrochemical cell consisting of $Zn^{2+}|Zn$ and $Cu^{2+}|Cu$ half-cells.

EXAM HINT

It is important that you can **describe** clearly how the charge flows in an electrochemical cell (such as the one shown in **fig A**).

Remember that it is the ions that flow in the salt bridge and not electrons. Make sure that you can explain why the K^+ ions move towards the copper half-cell and the nitrate ions move towards the zinc half-cell.

The relevant standard electrode potentials for the redox systems involved are:

$$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s) \quad E^{\ominus} = -0.76\,V$$
$$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s) \quad E^{\ominus} = +0.34\,V$$

The $E^{\ominus}$ value for the Zn^{2+} | Zn half-cell is the more negative, so the zinc electrode will be the negative electrode of the cell. When the cell is in operation, i.e. it is generating an electric current, electrons will flow through the external circuit from the zinc electrode to the copper electrode.

The reactions taking place under these conditions are shown below.

At the negative electrode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$.

At the positive electrode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$.

The overall cell reaction is:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

CELL DIAGRAMS

It is not always convenient to draw a diagram of the full apparatus for a cell. For simplicity, chemists use a shorthand notation to represent half-cells. The half-cell made from zinc ions and zinc metal is written as:

$Zn^{2+}(aq)$ | $Zn(s)$ $\quad E^{\ominus} = -0.76\,V$

The solid vertical line indicates a phase boundary, in this case between an aqueous phase, $Zn^{2+}(aq)$, and a solid phase, Zn(s).

Some other examples are:

$Mg^{2+}(aq)$ | $Mg(s)$ $\quad E^{\ominus} = -2.37\,V$

$Cu^{2+}(aq)$ | $Cu(s)$ $\quad E^{\ominus} = +0.34\,V$

$Al^{3+}(aq)$ | $Al(s)$ $\quad E^{\ominus} = -1.66\,V$

$Ag^{+}(aq)$ | $Ag(s)$ $\quad E^{\ominus} = +0.80\,V$

When the electrode, i.e. the electrical connection between the solution and the external circuit, is a piece of platinum foil, the following convention is used:

$Pt(s)$ | $Fe^{3+}(aq)$, $Fe^{2+}(aq)$ $\quad E^{\ominus} = +0.77\,V$

Notice that because there is no phase boundary between $Fe^{3+}(aq)$ and $Fe^{2+}(aq)$, a comma is used to separate them, not a solid vertical line.

The standard hydrogen electrode is represented as follows:

$$H^+(aq) \mid \tfrac{1}{2}H_2(g) \mid Pt(s)$$

These shorthand notations can now be used to represent a cell comprising two half-cells. Convention dictates two things:

1. The two reduced forms of the species are shown on the outside of the cell diagram.
2. The positive electrode is shown on the right-hand side of the cell diagram.

Applying these conventions produces the following cell diagram for a cell formed by combining the $Zn^{2+}(aq)$ | Zn(s) and the $Cu^{2+}(aq)$ | Cu(s) half-cells:

$$Zn(s) \mid Zn^{2+}(aq) \,\vdots\, Cu^{2+}(aq) \mid Cu(s)$$

The double vertical lines (⁞⁞) represent the salt bridge.

The emf, $E^{\ominus}_{cell}$, of this cell is simply the *difference* between the two standard electrode potentials of the two half-cells.

$$Zn(s) \mid Zn^{2+}(aq) \,\vdots\vdots\, Cu^{2+}(aq) \mid Cu(s)$$
$$E^{\ominus} = -0.76\,V \qquad E^{\ominus} = +0.34\,V$$

The difference between these two numbers is 1.10, so the emf of the cell is 1.10 V. To indicate that the right-hand electrode, i.e. the copper, is the positive electrode of the cell, the emf is given a positive (+) sign.

So, the complete cell diagram is:

$$Zn(s) \mid Zn^{2+}(aq) \,\vdots\vdots\, Cu^{2+}(aq) \mid Cu(s) \qquad E^{\ominus}_{cell} = +1.10\,V$$

LEARNING TIP

Note that the sign for the $E^{\ominus}$ of the $Zn^{2+}|Zn$ half-cell is still given as negative, despite the half-cell being written in reverse. As mentioned in **Section 16A.1**, the sign of a standard electrode potential remains the same no matter how the half-cell is represented.

DID YOU KNOW?

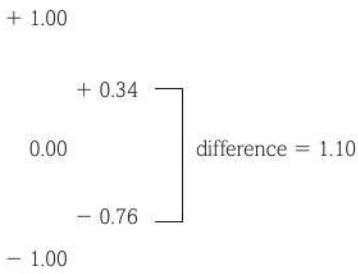
It is also possible to calculate the emf of the cell from a cell diagram by subtracting the $E^{\ominus}$ value of the left-hand half-cell from that of the right-hand half-cell:

$$E^{\ominus}_{cell} = E^{\ominus}_{right} - E^{\ominus}_{left}$$

It is important to remember, however, that you must not change the sign of the $E^{\ominus}$ value of the left-hand half-cell, even though the reaction is written as an oxidation and not a reduction.

WHAT IS MEANT BY THE 'DIFFERENCE' BETWEEN TWO STANDARD ELECTRODE POTENTIALS?

The easiest way to explain this is to represent the two values on a number scale, such as the one shown in **fig B**.



▲ **fig B** Example of a number scale to calculate an emf.

To move from one number to the other on the number scale involves a change of 1.10 units.

BREAKING THE CELL CONVENTION

Chemists allow themselves to break the rules if it suits their purpose. This is exactly what we do when we write cell diagrams to represent the measurement of a standard electrode potential. In this case, the standard hydrogen electrode is always written on the left-hand side.

The cell diagrams for the set-up used when measuring the standard electrode potentials of the $Zn^{2+}(aq) \mid Zn(s)$ and the $Cu^{2+}(aq) \mid Cu(s)$ half-cells are therefore:

$$Pt(s) \mid \tfrac{1}{2}H_2(g) \mid H^+(aq) \,\vdots\vdots\, Zn^{2+}(aq) \mid Zn(s) \qquad E^{\ominus}_{cell} = -0.76\,V$$

$$Pt(s) \mid \tfrac{1}{2}H_2(g) \mid H^+(aq) \,\vdots\vdots\, Cu^{2+}(aq) \mid Cu(s) \qquad E^{\ominus}_{cell} = +0.34\,V$$

As before, the sign of $E^{\ominus}_{cell}$ indicates the polarity of the right-hand electrode. Zinc is the negative electrode of the cell formed in combination with the standard hydrogen electrode. Copper is the positive electrode of the cell formed in combination with the standard hydrogen electrode.

CHECKPOINT

SKILLS CREATIVITY, ADAPTIVE LEARNING

1. (a) Draw a labelled diagram of the apparatus that can be used to construct a cell, under standard conditions, from a $Zn^{2+}|Zn$ half-cell and a $Fe^{3+}, Fe^{2+}|Pt$ half-cell.

(b) The standard electrode potentials for the half-cells are:

$Zn^{2+}|Zn \qquad E^{\ominus} = -0.76\,V$

$Fe^{3+}, Fe^{2+}|Pt \qquad E^{\ominus} = +0.77\,V$

Explain the direction of electron flow in the external circuit when this cell is in use.

(c) Write the cell diagram for this cell and calculate the emf ($E^{\ominus}_{cell}$) of the cell.

2. Chlorine may be prepared in the laboratory by reacting dilute hydrochloric acid with potassium manganate(VII). The standard electrode potentials that relate to this reaction are:

$\tfrac{1}{2}Cl_2(g) + e^- \rightleftharpoons Cl^-(aq) \qquad E^{\ominus} = +1.36\,V$

$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l) \qquad E^{\ominus} = +1.51\,V$

(a) Calculate the emf, $E^{\ominus}_{cell}$, of a cell constructed from these two redox systems.

(b) Explain the direction of electron flow that would take place in the external circuit of this cell when in use.

(c) Write the cell diagram for this cell.

16A 3 STANDARD ELECTRODE POTENTIALS AND THERMODYNAMIC FEASIBILITY

SPECIFICATION REFERENCE: 16.9 16.10 16.11 16.12 16.14

LEARNING OBJECTIVES

- Understand the importance of the conditions when measuring an electrode potential, *E*.
- Be able to use standard electrode potentials to predict the thermodynamic feasibility of a reaction.
- Understand that $E^{\ominus}_{cell}$ is directly proportional to the total entropy change and to ln*K* for a reaction.
- Understand the limitations of predictions made using standard electrode potentials, in terms of the kinetic stability of systems and departure from standard conditions.
- Understand how standard electrode potentials can be used to predict the thermodynamic feasibility of disproportionation reactions.

MAKING PREDICTIONS USING STANDARD ELECTRODE POTENTIALS

Using standard electrode potentials is one way of measuring how easily a species loses electrons. This method provides information on how far to the left an equilibrium is relative to the equilibrium in the standard hydrogen electrode. For example, for the following two equilibria, the equilibrium of the top reaction lies further to the left than that of the hydrogen electrode. Also, the equilibrium of the bottom reaction lies further to the right than that of the hydrogen electrode.

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ $\quad E^{\ominus} = -0.76\,V$

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $\quad E^{\ominus} = +0.34\,V$

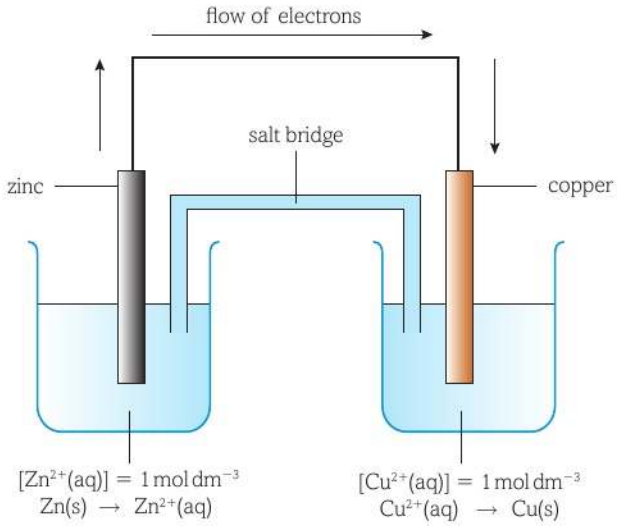
It follows, therefore, that the position of equilibrium of the $Zn^{2+}(aq) \mid Zn(s)$ reaction lies further to the left than that of the $Cu^{2+}(aq) \mid Cu(s)$ reaction.

If these two equilibria are linked by combining the two half-cells to make an electrochemical cell, then electrons will flow from the zinc electrode to the copper electrode. Therefore, the equilibria will be disturbed and the following reactions will take place in each half-cell:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$

Electrons flow from the half-cell with the more negative $E^{\ominus}$ value to the half-cell with the less negative (i.e. more positive) $E^{\ominus}$ value (**fig A**).



▲ **fig A** Electrochemical cell showing direction of flow of electrons.

LINKING THIS TO A TEST TUBE REACTION

Is the following reaction **thermodynamically feasible** (possible)?

$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$

That is, will zinc displace copper from an aqueous solution containing copper(II) ions?

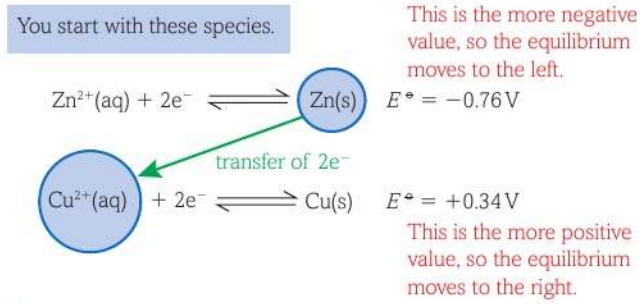
To answer this question we will look at the $E^{\ominus}$ values for the relevant half-cell reactions:

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ $\quad E^{\ominus} = -0.76\,V$ $\quad$ equilibrium 1

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $\quad E^{\ominus} = +0.34\,V$ $\quad$ equilibrium 2

The $E^{\ominus}$ value for equilibrium 1 is more negative than that for equilibrium 2. So, the position of equilibrium 1 will shift to the left, releasing electrons, and the position of equilibrium 2 will shift to the right, accepting electrons. This means that the reaction between zinc and copper(II) ions is thermodynamically feasible.

Fig B illustrates this process.

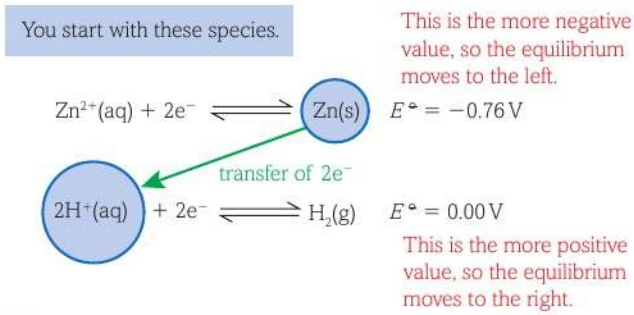


▲ **fig B** Reaction of zinc with copper(II) ions in aqueous solution.

WILL ZINC REACT WITH DILUTE SULFURIC ACID?

To answer this question we need to consider the following reaction:

$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$



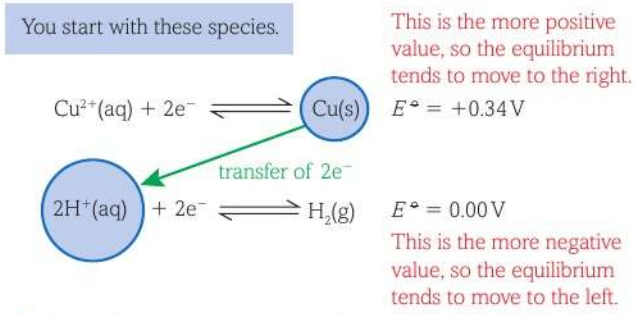
▲ **fig C** Reaction of zinc with dilute sulfuric acid.

As $E^\ominus[Zn^{2+}(aq) \mid Zn(s)]$ is more negative than $E^\ominus[2H^+(aq) \mid H_2(g)]$, the reaction is thermodynamically feasible, as zinc will release electrons to the hydrogen ions (**fig C**).

WILL COPPER REACT WITH DILUTE SULFURIC ACID?

To answer this question, we will consider the following reaction:

$$Cu(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2(g)$$



▲ **fig D** Copper does not react with dilute sulfuric acid.

This time, $E^\ominus[Cu^{2+}(aq) \mid Cu(s)]$ is more positive than $E^\ominus[2H^+(aq) \mid H_2(g)]$, so the reaction is not thermodynamically feasible because copper will not release electrons to hydrogen ions (**fig D**). The two equilibria tend to move in the opposite directions to those required for reaction.

Interestingly, the reverse reaction:

$$Cu^{2+}(aq) + H_2(g) \rightarrow Cu(s) + 2H^+(aq)$$

is thermodynamically feasible. However, no reaction takes place when hydrogen is bubbled into an aqueous solution containing copper(II) ions because the activation energy for the reaction is very large. The reactants are therefore **kinetically stable**.

AN ALTERNATIVE METHOD

There is an alternative method we can use to predict the thermodynamic feasibility of a reaction using standard electrode (redox) potentials. However, you need to understand that this method is based on imperfect science, since it involves reversing (i.e. changing back) one of the half-equations and changing the sign of the electrode potential. As we have already pointed out in **Section 16A.1**, it is incorrect to change the sign when reversing the half-equation. This is because the sign gives only the polarity of the half-cell when it is connected to a standard hydrogen electrode.

However, this alternative method does produce a correct answer regarding the feasibility of the reaction, despite the imperfect science. It involves calculating a value for $E^\ominus_{cell}$ for the proposed reaction. You should also understand that calculating a value for $E^\ominus_{cell}$ for a reaction is meaningless. This is because a reaction taking place in a reaction vessel, such as a test tube, does not generate an emf.

Below are two calculations using the alternative method.

WORKED EXAMPLE 1

Will zinc displace copper when added to an aqueous solution containing copper ions?

The proposed reaction is:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

The two standard electrode potentials to consider are:

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s)$ $\quad E^\ominus = -0.76\,V$

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $\quad E^\ominus = +0.34\,V$

To predict the feasibility of the reaction using this method, it is necessary to write the two half-equations as they occur in the proposed reaction, changing the sign of the standard electrode potential as necessary:

$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ $\quad E^\ominus = +0.76\,V$

$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ $\quad E^\ominus = +0.34\,V$

The best method is now to add together these two electrode potentials to produce an $E^\ominus_{cell}$ for the proposed reaction:

$$E^\ominus_{cell} = +0.76\,V + 0.34\,V = +1.10\,V$$

Because the value for $E^\ominus_{cell}$ is positive, the reaction is feasible.

WORKED EXAMPLE 2

Will copper produce hydrogen gas when added to dilute sulfuric acid?

The proposed reaction is:

$$Cu(s) + 2H^+(aq) \rightarrow Cu^{2+}(aq) + H_2(g)$$

The two electrode potentials to consider are:

$Cu^{2+}(aq) + 2e^- \rightleftharpoons Cu(s)$ $\quad E^\ominus = +0.34\,V$

$2H^+(aq) + 2e^- \rightleftharpoons H_2(g)$ $\quad E^\ominus = 0.00\,V$

Reversing the top half-equation and adding together the two electrode potentials gives:

$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ $\quad E^\ominus = -0.34\,V$

$2H^+(aq) + 2e^- \rightarrow H_2(g)$ $\quad E^\ominus = 0.00\,V$

and

$$E^\ominus_{cell} = -0.34\,V + 0.00\,V = -0.34\,V$$

Because the value for $E^\ominus_{cell}$ is negative, the reaction is not feasible.

You may be asked to calculate a value of $E^\ominus_{cell}$ to predict the feasibility of a reaction. We suggest that you learn this alternative method as well as the method described earlier that is based on correct science.

REACTION BETWEEN MANGANESE(IV) OXIDE AND HYDROCHLORIC ACID

One method of preparing chlorine in the laboratory is to react manganese(IV) oxide with hydrochloric acid:

$$MnO_2(s) + 4HCl(aq) \rightarrow Mn^{2+}(aq) + 2Cl^-(aq) + 2H_2O(l) + Cl_2(g)$$

However, the standard electrode potentials suggest that this reaction is not feasible.

$$MnO_2(s) + 4H^+(aq) + 2e^- \rightleftharpoons Mn^{2+}(aq) + 2H_2O(l) \quad E^\ominus = +1.23\,V \quad \text{equilibrium 1}$$

$$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-(aq) \quad E^\ominus = +1.36\,V \quad \text{equilibrium 2}$$

The $E^\ominus$ value for equilibrium 2 is more positive (i.e. less negative) than that for equilibrium 1. So, chloride ions cannot release electrons to MnO_2, and this is required for the reaction shown above to take place. The reaction is not thermodynamically feasible under standard conditions, in which the concentration of the hydrochloric acid, and therefore the concentration of both the hydrogen ions and chloride ions, is 1 mol dm^{-3}.

The key to making the reaction take place is to increase the concentration of hydrochloric acid.

If concentrated hydrochloric acid (~10 mol dm^{-3}) is used, then the concentration of both the hydrogen ions and the chloride ions increases. The effect is to shift the position of equilibrium 1 to the right and the position of equilibrium 2 to the left (**fig E**).

equilibrium shifts to the right →

$$MnO_2(s) + 4H^+(aq) + 2e^- \rightleftharpoons Mn^{2+}(aq) + 2H_2O(l)$$

$$Cl_2(g) + 2e^- \rightleftharpoons 2Cl^-(aq)$$

← equilibrium shifts to the left

▲ **fig E** Equilibria for the reaction of manganese(IV) oxide with hydrochloric acid.

As a result, the electrode potential, E, for equilibrium 1 becomes more positive because the redox system is now a better electron acceptor.

The electrode potential for equilibrium 2 becomes less positive because the redox system is now a better electron releaser.

The net effect is that the electrode potential of equilibrium 2 becomes less positive (more negative) than that of equilibrium 1, so the chloride ions can now release electrons to manganese(IV) oxide. The reaction has become thermodynamically feasible under these non-standard conditions.

SUMMARY

- The thermodynamic feasibility of a chemical reaction can be predicted using standard electrode potentials.
- Although the standard electrode potentials indicate that a reaction is thermodynamically feasible, it may not take place for two reasons:
  - the reactants may be kinetically stable because the activation energy for the reaction is very large, and
  - the reaction may not be taking place under standard conditions.
- A reaction that is not thermodynamically feasible under standard conditions may become feasible when the conditions are altered.
- Changing the conditions may alter the electrode potential, E, of a half-cell because the position of equilibrium of the half-cell reaction may change.

DISPROPORTIONATION REACTIONS

In **Topic 8 (Book 1: IAS)** we learned that a **disproportionation** reaction is one in which an element in a species is both oxidised and reduced in the same reaction at the same time.

A typical disproportionation reaction is the conversion of copper(I) ions in aqueous solution into copper(II) ions and copper atoms.

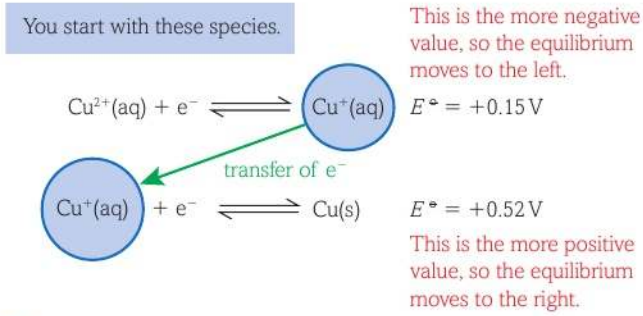
$$Cu^+(aq) + Cu^+(aq) \rightarrow Cu^{2+}(aq) + Cu(s)$$

This reaction can be explained in terms of standard electrode potentials.

$$Cu^{2+}(aq) + e^- \rightleftharpoons Cu^+(aq) \quad E^\ominus = +0.15\,V \quad \text{equilibrium 1}$$

$$Cu^+(aq) + e^- \rightleftharpoons Cu(s) \quad E^\ominus = +0.52\,V \quad \text{equilibrium 2}$$

The $E^\ominus$ value for equilibrium 1 is more negative than that for equilibrium 2, so the position of equilibrium 1 will move to the left, and the equilibrium position of 2 will move to the right. Therefore, Cu^+ ions will both release and accept electrons to one another to form Cu^{2+} ions and Cu atoms. That is, Cu^+ ions disproportionate into Cu^{2+} ions and Cu atoms (**fig F**).



▲ **fig F** Disproportionation of Cu^+ ions into Cu^{2+} ions and Cu atoms.

RELATIONSHIP BETWEEN TOTAL ENTROPY AND $E^\ominus_{CELL}$

The relationship between the total entropy change, $\Delta S^\ominus_{total}$, and the emf of the cell, $E^\ominus_{cell}$, is given by the following equation:

$$T\Delta S^\ominus_{total} = nFE^\ominus_{cell}$$

where n is the number of moles of electrons involved in the cell reaction and F is the Faraday constant.

As n is a constant for a given cell reaction, and F is also a constant, it follows that, at a given temperature, the total entropy change of the cell reaction is proportional to the emf of the cell.

$$\Delta S^\ominus_{total} \propto E^\ominus_{cell}$$

EXAM HINT

If you are going to use this equation in your answer you must indicate that it is $\Delta S^{\ominus}_{total}$ that is proportional to E_{cell} and not $\Delta S^{\ominus}_{system}$.

If $E^{\ominus}_{cell}$ is positive, the reaction as written from left to right in the cell diagram is thermodynamically feasible because $\Delta S^{\ominus}_{total}$ will be positive.

For example, for the following cell diagram:

$$Zn(s) \mid Zn^{2+}(aq) \,\vdots\, Cu^{2+}(aq) \mid Cu(s) \qquad E^{\ominus}_{cell} = +1.10\,V$$

the half-cell reactions that will take place in the cell are:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^- \text{ and } Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$$

giving an overall reaction of:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

In the following cell diagram:

$$Pt(s) \mid \tfrac{1}{2}H_2(g) \mid H^+(aq) \,\vdots\, Zn^{2+}(aq) \mid Zn(s) \qquad E^{\ominus}_{cell} = -0.76\,V$$

the half-cell reactions as written from left to right will not occur, as $E^{\ominus}_{cell}$ is negative. Instead the half-cell reactions will occur from right to left:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^- \text{ and } H^+(aq) + e^- \rightarrow \tfrac{1}{2}H_2(g)$$

giving an overall reaction of:

$$Zn(s) + 2H^+(aq) \rightarrow Zn^{2+}(aq) + H_2(g)$$

This means that $E^{\ominus}_{cell}$ can be used to predict the direction of a cell reaction.

RELATIONSHIP BETWEEN THE EQUILIBRIUM CONSTANT AND $E^{\ominus}_{CELL}$

The total entropy change for a reaction and the equilibrium constant, K, are related by the following equation:

$$\Delta S^{\ominus}_{total} = R \ln K$$

However:

$$\Delta S^{\ominus}_{total} = \frac{nFE^{\ominus}_{cell}}{T}$$

Therefore:

$$R \ln K = \frac{nFE^{\ominus}_{cell}}{T}$$

or:

$$\ln K = \frac{nFE^{\ominus}_{cell}}{RT}$$

As n, F and R are constants, it follows that at a given temperature, T, $\ln K$ is proportional to $E^{\ominus}_{cell}$:

$$\ln K \propto E^{\ominus}_{cell}$$

$E^{\ominus}_{cell}$ can therefore be used to calculate the thermodynamic equilibrium constant, K, for a cell reaction.

CHECKPOINT

SKILLS PROBLEM-SOLVING

1. (a) Use the following data to explain why, under standard conditions, VO^{2+} can be used to reduce MnO_4^- in acidic solution.

$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l) \quad E^{\ominus} = +1.00\,V$

$MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l) \quad E^{\ominus} = +1.51\,V$

(b) Write an ionic equation for the reaction that takes place.

2. Use the following data to explain to which oxidation state vanadium(V), in the form of VO_2^+, can be reduced by zinc.

$VO_2^+(aq) + 2H^+(aq) + e^- \rightleftharpoons VO^{2+}(aq) + H_2O(l) \quad E^{\ominus} = +1.00\,V$

$VO^{2+}(aq) + 2H^+(aq) + e^- \rightleftharpoons V^{3+}(aq) + H_2O(l) \quad E^{\ominus} = +0.34\,V$

$V^{3+}(aq) + e^- \rightleftharpoons V^{2+}(aq) \quad E^{\ominus} = -0.26\,V$

$V^{2+}(aq) + 2e^- \rightleftharpoons V(s) \quad E^{\ominus} = -1.18\,V$

$Zn^{2+}(aq) + 2e^- \rightleftharpoons Zn(s) \quad E^{\ominus} = -0.76\,V$

3. (a) Use the following data to explain why hydrogen peroxide, H_2O_2, decomposes into water and oxygen in aqueous solution.

$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightleftharpoons 2H_2O(l) \quad E^{\ominus} = +1.77\,V$

$O_2(g) + 2H^+(aq) + 2e^- \rightleftharpoons H_2O_2(l) \quad E^{\ominus} = +0.68\,V$

(b) Write an equation for this decomposition.

(c) Explain, using oxidation numbers, why this reaction is classified as disproportionation.

4. $Fe^{3+}(aq)$ ions can be reduced by $I^-(aq)$ ions. The equation for the reaction is:

$Fe^{3+}(aq) + I^-(aq) \rightarrow Fe^{2+}(aq) + \tfrac{1}{2}I_2(aq)$

The standard electrode potentials for $Fe^{3+}(aq)$, $Fe^{2+}(aq)$ and $\tfrac{1}{2}I_2(aq)$, $I^-(aq)$ are +0.77 V and +0.54 V, respectively.

[The gas constant, R, is 8.31 J mol^{-1} K^{-1}. The Faraday constant, F, = 96 500 C mol^{-1}.]

(a) Calculate the emf of a cell made by combining these two half-cells.

(b) Calculate the value of the equilibrium constant, K, for the cell reaction.

SUBJECT VOCABULARY

thermodynamically feasible reaction a reaction that should take place without any intervention by us, if we consider the enthalpy and entropy changes involved

kinetically stable the reaction does not take place, or is very slow, because the activation energy for the reaction is very high

disproportionation a reaction in which an element is both oxidised and reduced at the same time